# Wearable Pipeline: Data, Training, Inference and Overlap

Code-reading notes: 4 November 2026

This describes the current implementation, rather than assuming every statement in the older documentation remains current. No project code or tests were run for this review. The concerns below come from static inspection and have not been reproduced in a fresh experiment.

## 1 The main distinction

The pipeline has two kinds of learning: estimating each patient's signal behaviour, and training a classifier to recognise exacerbations. They happen at different stages.

Raw recordings and clinical labels
↓
Separate channel files and a daily label table
↓
Raw / smoothing / frozen forecasting / frozen encoding
↓
Optional change-point detection
↓
Daily features and history windows
↓
Patient-level train/test split
↓
Supervised classifier $\longrightarrow$ held-out probabilities and evaluation

The signal-processing routes are alternative experimental branches. They do not all have to run consecutively.

## 2 Raw data to channel files and labels

The default source reads `data/raw/pass1/patients.csv`, selects patients marked `in_models`, and loads their minute-level recordings.

- Clinical labels are collapsed to one row per patient-day before filtering wearable readings. The labels already exist in the export; this repository reads them.
- HRV, heart rate, temperature, steps, sleep and activity channels are extracted separately. Each channel retains its own observation times.

- Long gaps receive NaN filler rows. These mark missingness without inventing readings. Filtering chunks are defined from actual observations, with a gap of at least 180 minutes opening a new chunk.

Nothing is trained here. Output is one processed file per patient-channel and a separate daily ground-truth table, `data/processed/labels_pass1.csv`.

**Code:** `wearable/sources.py`; `wearable/steps/ingest.py`; `wearable/chunking.py`.

## 3 What each signal-processing route learns

| Route | What is fitted or trained? | Output |
|---|---|---|
| Raw | Nothing. | Original readings. |
| Smoothing | Patient-specific parameters and evolving filter states; no clinical labels. | Baseline and residual: raw minus baseline. |
| Frozen forecasting | Timestral, TimesFM or Chronos pretrained weights remain frozen. | Predicted readings, forecast error (surprise), and distribution position (tail). |
| Frozen encoding | MOMENT or Mantis pretrained weights remain frozen. Each input window is normalised separately. | One representation vector per patient-day. |

**The frozen forecasters predict physiological readings, not exacerbations.** Their outputs become features for a separate clinical classifier. Forecasting uses preceding observations to predict blocks ahead. Encoding instead reads a window including the day being classified and summarises it as a vector.

The experiment normally processes HRV and HR separately. Other available channels join later as raw features. Chronos is currently used as a univariate forecaster; its possible cross-channel covariate route is not implemented.

**Code:** `wearable/experiment.py` (`inputs`); `wearable/steps/smooth.py`; `wearable/steps/forecast.py`; `wearable/steps/encode.py`.

## 4 Optional change-point detection

Change-point detection (CPD) can read raw values, smoothed values, smoothing residuals, forecast surprise, or forecast tail. For each patient and channel it:

1. Estimates an hour-of-day and behavioural-state baseline from prior readings.
2. Converts deviations into standardised innovations using a noise model.
3. Calibrates detector thresholds using synthetic no-change histories built from that patient's past.
4. Applies BOCPD, KCPD, Kalman, HMM and nightly drift detection.
5. Combines detector alarms using OR.

This is patient-specific, unsupervised adaptation, including on test patients' own histories. It does not use exacerbation labels. Default calibration shuffles historical days; optional STS calibration fits a generative time-series model to past readings. The noise model refits from trailing history, and alarm thresholds are recalibrated periodically.

The output contains flags, magnitudes, noise statistics and gap-reentry information. It is not a probability of exacerbation. Causal detector processing also cannot undo future leakage already introduced by its input smoother.

**Code:** `wearable/steps/detect.py`; `wearable/cpd/_whiten.py`; `wearable/cpd/_sts.py`.

## 5 Readings to classifier examples

The `join` step builds one example per patient-day:

- Continuous readings become daily mean, standard deviation, minimum and maximum.
- State/activity and alarm columns contribute totals and bout counts.
- Optional profiles preserve time-of-day bins instead of only daily summaries.
- A seven-day input window attaches day $D$ and days $D-1$ through $D-6$. Missing calendar days remain missing; older days do not shift into their places.
- Embeddings already represent the requested window, so they are not lagged again.
- Clinical labels are attached by patient and date. Identifiers, clinical labels, event IDs and cohort group are excluded from classifier features.

| Task | Available features | Target |
|---|---|---|
| Detection | Through day $D$, including $D$. | Is $D$ an exacerbation day? |
| Onset | Through day $D$, including $D$. | Does a new episode start within the next $H$ days? |

The tasks produce separate datasets and results. The default onset horizon is three days; it can be narrowed to one or two while retaining the same eligible days.

**These are full-day predictions.** Using all readings from $D$ supports a decision after those readings are available. It does not establish an alert available at the beginning of $D$.

The default eligibility rule retains the study's HRV-based sample days even when other channels are included as features. An optional wider validity rule exists; changing eligibility changes the population and needs separate review.

**Code:** `wearable/steps/join.py`; `wearable/windowing.py`.

## 6 Supervised training and held-out inference

Patients are split 80:20 by default, stratified by cohort and whether they have an exacerbation. All days of a patient stay on one side. Five seeds repeat the experiment. Within a comparison, input methods use the same split.

The final model receives training features and training labels:

- **XGBoost**, the default, learns boosted trees; CatBoost is another tree classifier. Both handle class imbalance during training.
- **RNN/LSTM** learn sequences of daily feature vectors. Feature scales are estimated from training data and reused for test inference.
- **Mantis classifier** uses pretrained Mantis and trains it for the clinical task. The default requests full fine-tuning, but when a channel adapter is required, the implementation switches to adapter/head training.
- **Always-no** is the reference that never flags a day.

The fitted model then predicts $P(y = 1)$ for held-out patients. It does not receive their clinical labels during prediction. Evaluation subsequently compares probabilities with those labels, computes metrics, resamples test patients for confidence intervals, and writes probability-band calibration tables.

**The calibration table assesses probabilities; it does not recalibrate them.** It is also distinct from the unsupervised alarm-threshold calibration in CPD.

The model step saves predictions and provenance, not a fitted classifier checkpoint. It is currently an experiment workflow rather than a separate saved-model deployment flow. The artifact store caches results so the same signal-processing outputs can be reused across tasks, models and folds.

**Code:** `wearable/steps/model.py`; `wearable/models/`; `wearable/steps/evaluate.py`; `wearable/store.py`.

# 7 Where the data and features overlap

1. **Adjacent days.** Seven-day windows overlap by six days within a patient. This is expected. Patient-level splitting prevents these windows from crossing train/test within a run.
2. **Different seeds.** A patient may train in one run and test in another. These are repeated holdouts, not five independent cohorts.
3. **Different branches.** A smoothed artifact carries raw values, baseline and residual. A residual-plus-CPD route passes the smoothing block and detector features. An embedding route adds embeddings beside ordinary raw features. Therefore these comparisons concern feature bundles, not isolated representations.
4. **Before the split.** Signal processing runs before the patient split and is reused across folds. Frozen inference and adaptation using a test patient's available past can be legitimate. Using that patient's future is not causal.
5. **Signal-only smoothing studies.** The separate smoothing-study script can derive settings from entire cohorts. If those settings are later used for a held-out predictive claim, their derivation must also respect the training boundary; being label-free alone does not establish an untouched test set.

# 8 Concrete concerns to discuss before corrections

## 8.1 Future-dependent smoother calibration

Nine of the ten smoothers estimate patient-specific parameters from the full supplied recording. Later observations can therefore change earlier features. Patient-level splitting does not fix this

within-patient lookahead. The documented truncation-exact exception is `anrewma`; this review did not rerun that check.

For example, `gammadglm` fits its patient calibration before filtering individual chunks. A forward filtering loop does not make the whole operation causal when its parameters were estimated using future readings.

### 8.2 CPD channel bookkeeping

Detect artifacts do not record `stats.signal`, but `join._features` uses that field to decide which raw channel an artifact already covers. The fallback is HRV. An HR detector can therefore be treated as covering HRV, changing which raw feature blocks are included. This needs correction or explicit verification before assuming the intended variables reach every path.

### 8.3 Flag counts labelled as minutes

Daily aggregation sums flagged rows into columns ending in `_minutes`. For irregular HRV readings, a flagged-row count is not elapsed time. The feature should either be named as a count or computed using an explicit duration rule.

## 9 Scope of these notes

These notes explain the implemented flow and identify specific points for discussion. They do not certify causality, vendor-model behaviour, clinical performance, or existing cached results. No changes to pipeline code were made, and no experiments, tests or LaTeX compilation were run for this document.